

A High-Precision Phenomenological Relation for Newton's Constant with Exact $B_3/SO(7)/SO(8)$ Algebraic Scaffold and an Explicit Mechanism Gap

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Public proof package: <https://github.com/oldrich-research/gravitational-constant-relation>

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Abstract

The relation

$$G = \frac{4}{3} \frac{\hbar c}{m_e^2} \alpha^{21} \exp\left(-\frac{5\alpha}{2}\right)$$

evaluates on CODATA 2022 inputs [1] to $G_{\text{pred}} = 6.6743123 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, residual $+1.84 \text{ ppm}$ ($z = +0.082\sigma$). Against the April 2026 NIST/Metrologia single-experiment result [10], residual $+66.3 \text{ ppm}$ ($+1.16\sigma$). Propagated input uncertainty: 3.3×10^{-9} , four orders below the experimental floor.

The discrete coefficients identify exactly with $B_3/SO(7)/SO(8)$ Lie-algebraic data: $21 = \dim(\text{adj } SO(7))$; $5/2 = h^\vee(B_3)/2$; $4/3 = \dim(SO(8))/\dim(SO(7))$.

A bounded reproducibility scan over 22,680 candidates of class $A(\hbar c/m_e^2)\alpha^n \exp(-k\alpha)$ places $(4/3, 21, 5/2)$ at rank 1 within 0.5σ and 0.1σ ; nearest competitor -3640 ppm at -162σ . A Julia bounded-equation-search engine enumerates 2.02×10^9 raw expressions (2.46×10^6 canonical-unique) through cost shell $C \leq 20$; with `alpha_family` pre-seeded, no competitor beats the canonical formula at budgets 30 and 40. Recovered rank-1 `rel_err` = 1.85×10^{-6} , invariant across $C = 18, 19, 20$. Sham-parity: 0/256 at $C = 18$, 1/256 at $C = 19, 20$ (single non- α TRASH expression).

A structured survey across QED, soft-photon and Sudakov resummation, instantons, functional determinants on S^7 and B_3 , heat-kernel localization, one-loop β -coefficients, $SO(7)^\pm$ gauged supergravity, induced gravity, and the Kalinski QED-vacuum kernel produces no mechanism reproducing the α^{21} counting, $\exp(-5\alpha/2)$ damping, and m_e normalization simultaneously. The relation is a high-precision phenomenological anomaly with an exact algebraic scaffold and a documented mechanism gap. Operational falsification exposure: $\sim 13.25 \text{ ppm}$ at 5σ against any future low-uncertainty G recommendation converging on the NIST 2026 central value.

AI authorship. This paper reports research carried out by an orchestration of AI agents under the named author's direction. The author is an independent researcher without specialist training in theoretical physics; his role was vision, target selection, and approval. See *Statement of AI Authorship* for the full disclosure.

Headline Quantities

1. The Relation and Its Numerical Status

The relation evaluates to $G_{\text{pred}} = 6.6743123 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ on CODATA 2022 inputs [1]. Against the recommended value $G = 6.67430(15) \times 10^{-11}$ this is $+1.84 \text{ ppm}$ and $z = +0.082\sigma$. The integer exponent required to hit the CODATA 2022 central value exactly is $n_{\text{exact}} = 21.00000037$.

Cross-benchmark status against the April 16, 2026 NIST/Metrologia single-experiment determination $G = 6.67387(38) \times 10^{-11}$ [10] is $+66.3 \text{ ppm}$ and $z = +1.16\sigma$.

Propagation of CODATA input uncertainties gives a relative uncertainty in G_{pred} of 3.3×10^{-9} , dominated 96.5% by α and 3.5% by m_e . This is below the experimental uncertainty on G by a factor of 6.9×10^3 . $\hbar$ and c are exact by SI definition. Re-computation across successive CODATA releases gives $z = +0.75\sigma$ (2014), $+0.082\sigma$ (2018), $+0.082\sigma$ (2022): the 2014→2018 improvement was driven by movement of the experimental central value toward the formula.

2. Bounded-Family Reproducibility Search

The search class is

$$G = A \frac{\hbar c}{M^2} \alpha^n \exp(-k\alpha)$$

with declared parameters fixed as follows:

- Mass scale $M = m_e$ (CODATA 2022 [1]).
- Exponent window $n \in \{5, 6, \dots, 40\}$ (36 values).
- Damping window $k \in \{0, 1/2, 1, \dots, 10\}$ (21 half-integer values).
- Prefactor prior: the reduced-rational set generated by p/q with $p \leq 5$, $q \leq 30$, $\text{gcd}(p, q) = 1$,

$p/q \in [1/3, 5]$. Explicit list (30 values):

$$\left\{ \frac{1}{3}, \frac{5}{14}, \frac{4}{11}, \frac{3}{8}, \frac{5}{13}, \frac{2}{5}, \frac{5}{12}, \frac{3}{7}, \frac{4}{9}, \frac{5}{11}, \frac{1}{2}, \frac{5}{9}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{5}{7}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, 1, \frac{5}{4}, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, 2, \frac{5}{2}, 3, 4, 5 \right\}.$$

- Total cardinality: $30 \times 36 \times 21 = 22,680$ candidates.
- Canonical candidate-list SHA-256:
37ee945afb33e5c8f4a63b9874338fb3
8d8cab1e8b1d2c113873a2e632ff92f2.

Within this declared family, the canonical $(A, n, k) = (4/3, 21, 5/2)$ is the unique candidate within 0.5σ and the unique candidate within 0.1σ of the CODATA 2022 central value of G [1]. It is also rank 1 against the NIST 2026 central value [10].

The five nearest-neighbor rows against CODATA 2022 [1]:

The same five rows against NIST 2026 [10] preserve rank order; canonical residual $+66.3 \text{ ppm}$, rank-2 residual -3576 ppm . Arithmetic rank-loss midpoints between rank 1 and the immediate k -neighbors $(4/3, 21, 3)$ and $(4/3, 21, 2)$ sit at -1821.01 ppm and $+1827.67 \text{ ppm}$ relative to G_{pred} . The 66.3 ppm NIST offset is two orders of magnitude smaller than that rank-loss scale.

Replacing m_e by m_p in the same scan produces no candidate within either 0.5σ or 0.1σ of the CODATA 2022 central value.

The reduced-rational prior with $p \leq 5$, $q \leq 30$, $p/q \in [1/3, 5]$ is the publication declaration. Reproduction one-liner:

```
python3 scan.py --benchmark
codata2022 --limit 5
```

bundled in the public proof package; expected output reproduces the candidate-list SHA-256 above and the top-five table.

The bounded scan certifies uniqueness within the declared functional class with m_e fixed. It does not certify uniqueness against arbitrary enlargements of the class; the look-elsewhere problem outside the

Quantity	Value
G_{pred} and residuals	$6.6743123 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$; +1.84 ppm at $+0.082\sigma$ vs CODATA 2022 [1]; +66.3 ppm at $+1.16\sigma$ vs NIST 2026 [10]
Bounded prior scan (22,680 candidates)	canonical (4/3, 21, 5/2) at rank 1 within both 0.5σ and 0.1σ ; nearest non-canonical row -3640 ppm at -162σ
AV6 symbolic engine, $C \leq 20$	2.02×10^9 raw, 2.46×10^6 canonical-unique; canonical rank 1 at budgets 30, 40; $\text{rel_err} = 1.85 \times 10^{-6}$ invariant across $C=18, 19, 20$
Sham-parity check (256 targets, budget 40)	0/256 at $C=18$; 1/256 at $C=19, 20$ (non- α TRASH)
Exact algebraic identities	$21 = \dim(\text{adj } SO(7))$; $5/2 = h^\vee(B_3)/2 = \rho \cdot e_1$; $4/3 = \dim(SO(8))/\dim(SO(7))$
Falsification threshold	~ 13.25 ppm at 5σ

Rank	A	n	k	$G_{\text{pred}} \times 10^{11}$	Residual (ppm)	z -score
1	4/3	21	5/2	6.6743123	+1.84	+0.082
2	4/3	21	3	6.6500043	-3640.19	-161.97
3	4/3	21	2	6.6987092	+3657.19	+162.73
4	4/3	21	7/2	6.6257847	-7268.96	-323.44
5	4/3	21	3/2	6.7231952	+7325.90	+325.97

declared boundary is treated by the Julia bounded-equation-search engine of Section 5 (audit row AV6).

3. Algebraic Scaffold

The B_3 positive root system contains nine roots: three short positive roots e_i for $i = 1, 2, 3$, and six long positive roots $e_i \pm e_j$ for $1 \leq i < j \leq 3$. The Weyl vector $\rho = \frac{1}{2} \sum_{\alpha > 0} \alpha = (5/2, 3/2, 1/2)$ in the standard orthonormal basis. The basis-invariant statement is $\rho \cdot e_1 = h^\vee(B_3)/2$; the numerical value $5/2$ follows in the standard dominant-chamber basis. The dual Coxeter number is $h^\vee(B_3) = 5$.

The coefficient $5/2$ is not $C_2(\text{vector}, SO(7))$. In the standard normalization $C_2(\text{vector}, SO(7)) = 3$; the correct identification is $5/2 = h^\vee(B_3)/2 = \rho \cdot e_1$. Two further B_3 arithmetic equalities also give $4/3$ — $(h^\vee(B_3) - 1)/\text{rank}(B_3) = (5 - 1)/3 = 4/3$ and $(\text{rank}(B_3) + 1)/\text{rank}(B_3) = (3 + 1)/3 = 4/3$ — but these are not independent invariant derivations; the primary anchor for $4/3$ is the $SO(8)/SO(7)$ embedding ratio.

The scaffold provides exact identifications for the discrete content of the formula. It does not provide a dynamical derivation. Section 4 documents that

no surveyed mechanism class closes the dictionary from the scaffold to the physical appearance of α , m_e , and the exponential dressing.

4. Mechanism Gap: Surveyed Routes

The following derivation classes were surveyed and did not produce a mechanism reproducing the full formula. Each row lists the expected contribution under that class, the observed surveyed output, and the structural obstruction.

The three irreducible anchors a future microscopic theory would need to close simultaneously:

1. α^{21} **counting**. A physical state space or dynamical system that produces 21 independent modes mapped onto the $SO(7)$ adjoint, rather than analogizing after the fact. The orbit-dimension ceiling $18 < 21$ rules out localization on B_3 along any root ray.
2. $\exp(-5\alpha/2)$ **damping**. A dynamical process producing a constant exponential factor with α in the numerator at $\alpha = \alpha_{\text{EM}}$. Surveyed semiclassical actions place α in the denominator; perturbative expansions give polynomials in α ; soft resummation gives logarithms in

Coefficient	Identification	Source
21	$\dim(\text{adj } SO(7)) = 7 \cdot 6/2$	exact
5/2	$\rho \cdot e_1 = h^\vee(B_3)/2$, dominant-chamber convention	exact, Humphreys [3]
4/3	$e_1 = \omega_1$ $\dim(SO(8))/\dim(SO(7)) = 28/21$	exact (primary anchor)

Route class	Expected contribution	Surveyed output	Obstruction
Heat-kernel / Weyl-character / localization on B_3	α^{21} counting from B_3 -orbit data	$\alpha^5 \exp(-5\alpha/2)$ at leading order along e_1	Active-root count along e_1 is 5; $SO(7)$ adjoint real-orbit dimension ceiling is $18 < 21$
Instanton / Yang-Mills saddle	Finite exponential dressing $\exp(-c\alpha)$	$\exp(-c/\alpha)$	Semiclassical action places α in the denominator
Soft-photon resummation (YFS / Sudakov / threshold)	Linear- α exponentiation	$\exp(\alpha \cdot C \cdot \log)$, energy-scale logs	Logarithms in energy scale instead of constant $\exp(-5\alpha/2)$
Functional determinant on S^7	Link S^7 determinant to $\alpha^{21} \exp(-5\alpha/2)$	No gauge zero-mode support; wrong determinant size and sign	$H^1(S^7) = 0$ removes zero-mode structure
One-loop β -function / threshold coefficient $SO(7)^\pm$ gauged-supergravity vacua	5/2 from standard running $SO(7)$ adjoint structure from $N=8$ supergravity	$\beta_0 = 55/3$ (pure YM, B_3); $\beta_0 = 15$ ($N=1$ SYM, B_3) Both $SO(7)^\pm$ vacua unstable; stable $N=1$ vacuum is G_2 , not $SO(7)$	Neither standard normalization equals 5/2 de Wit / Nicolai [4]; Borghese et al. [5]; Fischbacher [6]
Induced gravity / vacuum response	Electron-mass normalization from vacuum response	Sakharov-class arguments do not single out m_e	Visser [7]; Chaichian et al. [8]
Kalinski QED-vacuum kernel	Electron-scale normalization	Bare kernel $(4/3)(\hbar c/m_e^2)\alpha^{21}$ with fitted $2^{-1/38}$ factor	No $B_3/SO(7)$ -anchored derivation of $\exp(-5\alpha/2)$ [9]

energy scale.

- Electron-mass normalization.** A reason why $\hbar c/m_e^2$ is the gravitational reference scale. The m_p -substituted bounded scan produces no comparable hit. The same locked $B_3/SO(7)$ algebraic scaffold also fails a pre-registered lepton mass-ratio holdout: no parameter-free candidate reproduces both m_μ/m_e and m_τ/m_e , and the best parameter-free muon-ratio near-hit misses by 1408 ppm ($\sim 760\times$ worse than the G residual).

The exclusion is over surveyed standard classes, not over all logically possible theories.

5. Adversarial Validation Audit

The relation was attacked along seven independent stress modes. Each row is conditional on the

declared bounded prior and engine configuration cited.

AV6 Engine Compute Trail

The Julia bounded-equation-search engine (`run_av6_exhaustive_c40_jl15.jl`, source in the public proof package) implements grouped/cross-merge canonical reduction, weighted dynamic chunking, and per-target sham-parity bookkeeping. Canonical reduction performs in-enumeration equality saturation: expressions are quotiented modulo associativity and commutativity of $+$, $\cdot$, multiplicative inverse identities, α -degree normalization, and exponent collection, so the 2.46×10^6 canonical-unique count at $C=20$ is a distinct-expression count, not raw chunk output. Idempotence of the canonical form and hash stability across re-runs are checked at

ID	Stress mode	Declared scope	Result
AV0	Frozen prior	22,680-candidate prior, declared A, n, k menus, candidate-list SHA-256 fixed	Canonical (4/3, 21, 5/2) rank 1; two independent code paths reproduce list and residual
AV1	Sham-target tribunal	$N=10,000$ synthetic targets under AV0 grammar	True target rank 7/10001 on residual, 4/10001 under strict-cap comparator; surviving competitors concentrate on the $n=21$ shelf
AV2	Source/metrology ablation	Inputs perturbed within reported uncertainties inside AV0 family	Canonical row shifts ≤ 0.017 ppm / $\leq 0.0008\sigma$; rank and hit counts preserved
AV3	Scaffold invariant census	Frozen-complexity invariant scoring across tested single-algebra sets	$B_3/SO(7)$ canonical scaffold ranks 1
AV4	Blind scaffold prior audit	Independent blind algebraic scoring across alternative single-algebra scaffolds	$B_3/SO(7)$ remains top single-algebra scaffold; 4/3 arithmetic echoes recorded as Section 3 echoes
AV5	Lepton mass-ratio holdout	Pre-registered m_μ/m_e and m_τ/m_e under locked $B_3/SO(7)$ scaffold; 18-framework survey	Ruled out; best parameter-free muon-ratio near-hit 1408 ppm; Koide [2] charged-lepton relation (2/3) remains descriptor only
AV6	Bounded/frontier symbolic null	Julia bounded-equation-search engine with canonical <code>alpha_</code> family pre-seeded; cost shells $C=18, 19, 20$; evaluation budgets $\{20, 30, 40\}$; 256-target sham-parity	Rank-1 survival under adversarial competition: among 2.02×10^9 raw expressions (2.46×10^6 canonical-unique) at $C \leq 20$, the canonical formula is rank 1 at budgets 30 and 40 with $\text{rel_err} = 1.85 \times 10^{-6}$ invariant across $C=18, 19, 20$ (canonical internal engine cost 28, not evaluable at budget 20; budget-20 leader is a non- α pow at $\text{rel_err} \sim 10^{-4}$). Sham parity 0/256 at $C=18$, 1/256 at $C=19, 20$ (non- α TRASH at sham-index 28).
AV7	CODATA/NIST target stability	Same frozen prior re-evaluated against CODATA 2022 [1] and NIST 2026 [10] central values	Rank 1 against both targets; rank-loss midpoints -1821 ppm and $+1828$ ppm; ~ 13.25 ppm 5σ falsification width

engine startup; the 256-target sham-parity probe doubles as an adversarial engine-level test against accidental matches. Per-cost shell statistics at completion:

With the canonical `alpha_family` pre-seeded into the engine grammar, no competitor among the 2.02×10^9 raw expressions (2.46×10^6 canonical-unique after grammar reduction) at $C \leq 20$ matches or beats the canonical $(4/3)\alpha^{21} \exp(-5\alpha/2)$ at evaluation budgets 30 and 40. The 256-target sham-parity probe at the full $C \leq 20$ enumeration returns 1/256 — a single non- α TRASH pow expression — for a per-target hit rate $\sim 4 \times 10^{-3}$ on the observed CODATA residual under the same grammar and cost class; the $C=18$ publication baseline returns 0/256 at the same threshold.

The single sham drift to 1/256 at $C \geq 19$ is the same pow expression at `sham_index = 28`: $(\exp((5) - ((4)^2) \cdot \pi))^3$ at $C = 19$ and $(\exp((1) - (3 \cdot 4) \cdot \pi))^3$ at $C=20$, both classified as `TRASH_BROAD_GRAMMAR` and carrying no α dependence.

Budget-20 evaluations do not place the canonical at rank 1 because its internal engine cost is 28 and exceeds the budget cap; the canonical is not assigned a target-fit score under that constraint. The budget-20 leader is a non- α pow expression with $\text{rel_err} \sim 4.6 \times 10^{-4}$ at $C=18$ and $\sim 1.06 \times 10^{-4}$ at $C=19, 20$. At budgets 30 and 40 the canonical evaluates and dominates by three orders of magnitude in rel_err .

$C=18$ is the publication baseline; $C=19$ and $C=20$ extend sibling coverage and sham-parity robustness while preserving the same canonical rank-1 outcome and rel_err .

The j115 engine supersedes two earlier internal runners. The intermediate j114 had an `alpha_family` seed envelope that excluded the cost-28 canonical from the generated set; the omission was caught during pre-publication ledger audit and corrected in j115 (seed envelope raised to $\max(\text{budgets})$). The earlier j113 used a coarser sibling-class classifier and is not part of the reported ledgers. Both prior runners are present in the engine source tree for inspection; their outputs are not aggregated with the $C=18, 19, 20$ j115 ledgers reported

here. Detection of the j114 omission during audit is itself a positive datum on engine adversariality: the canonical was not granted to the engine by construction; it had to survive a generation-side seeding bug that initially hid it.

Three further audit paths were considered but not executed because they do not change the verdict at first order. (i) Extending shell completion beyond $C=20$ roughly doubles wall time per shell ($C=18 \sim 4.5$ h, $C=19 \sim 14$ h, $C=20 \sim 28$ h on 24 threads); given canonical rank 1 with identical $\text{rel_err} = 1.85 \times 10^{-6}$ is already invariant across $C=18, 19, 20$, additional shells would extend sibling coverage and sham-parity robustness rather than flip the outcome. (ii) Broadening the grammar with non- α primitives enlarges the sham pool but cannot remove a pre-seeded canonical from rank 1 since comparison is by absolute residual. (iii) Removing the `alpha_family` pre-seed tests whether brute-force enumeration alone reaches a cost-28 canonical within available compute, a different question than adversarial competition against a present canonical; the reported $C=18, 19, 20$ enumeration already answers the cost-coverage question through the explicit budget-20 versus budget-30/40 contrast.

6. Metrology Residuals

Predicted $G_{\text{pred}} = 6.6743123 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ compared against modern precision measurements and adjustments:

Sign convention: $\text{residual} = (G_{\text{pred}} - G) \times 10^6 / G$ in ppm; $z = (G_{\text{pred}} - G) / \sigma$. Positive entries indicate the formula prediction lies above the measured central value.

The CODATA adjustments aggregate the primary measurements through least-squares treatment. The spread among primary determinations (BIPM 2001 / 2013 on the high side, JILA-10 / Rosi on the low side) is visible in the residual column.

7. Prior Art Boundary

The public Rochester homepage of M. Kalinski [9] contains the bare kernel $(4/3)(\hbar c / m_e^2) \alpha^{21}$ with a numerically fitted correction factor

Shell completion C	Wall time (24 threads)	Cumulative raw	Cumulative canonical-unique	sham_match_or_beat at budget 40
18 (publication baseline, 2026-05-08)	~ 4.5 h	3.11×10^8	1.30×10^6	0/256
19 (extending evidence, 2026-05-09)	~ 14 h	8.17×10^8	1.80×10^6	1/256
20 (extending evidence, 2026-05-10)	~ 28 h	2.02×10^9	2.46×10^6	1/256

Source	Citation	$G \times 10^{11}$	$\sigma \times 10^{11}$	Type	Method	Residual (ppm)	z -score
CODATA 2014	Mohr et al. (2016)	6.67408	0.00031	Adjustment	LSQ	+34.80	+0.749
CODATA 2018	Tiesinga et al. (2021)	6.67430	0.00015	Adjustment	LSQ	+1.84	+0.082
CODATA 2022	Mohr et al. (2025) [1]	6.67430	0.00015	Adjustment	LSQ	+1.84	+0.082
UWash	Gundlach & Merkowitz (2000)	6.674215	0.000092	Primary	Torsion balance, AAF	+14.58	+1.057
BIPM 2001	Quinn et al. (2001)	6.67559	0.00027	Primary	Torsion strip, electrostatic	-191.40	-4.732
UZur-06	Schlamming et al. (2006)	6.67425	0.00012	Primary	Beam balance	+9.33	+0.519
JILA-10	Parks & Faller (2010)	6.67234	0.00014	Primary	Fabry-Perot	+295.59	+14.088
BIPM 2013	Quinn et al. (2013)	6.67554	0.00016	Primary	Torsion strip + TOS	-183.92	-7.673
Atom interferometry	Rosi et al. (2014)	6.67191	0.00099	Primary	Atom interferometry	+360.06	+2.427
UCI-14	Newman et al. (2014)	6.67433	0.00013	Primary	Torsion balance, TOS	-2.65	-0.136
HUST-18 TOS	Li et al. (2018)	6.674184	0.000078	Primary	Torsion balance, TOS	+19.22	+1.645
HUST-18 AAF	Li et al. (2018)	6.674484	0.000078	Primary	Torsion balance, AAF	-25.72	-2.201
NIST 2026	Schlamming et al. (2026) [10]	6.67387	0.00038	Primary	BIPM torsion at NIST	+66.27	+1.164

$2^{-1/38}$. An archived snapshot of that page and a bundled local capture are part of the public proof package (`public_proof_package/kalinski_rochester_capture_2026-04-23.md`).

The present manuscript adds:

1. The $B_3/SO(7)/SO(8)$ algebraic scaffold for the discrete coefficients 21, $5/2$, and $4/3$.
2. The replacement of the fitted $2^{-1/38}$ factor by the B_3 -anchored $\exp(-5\alpha/2)$, tied to $h^\vee(B_3)/2 = \rho \cdot e_1$.
3. The bounded 22,680-candidate reproducibility scan with declared prior and candidate-list SHA-256.
4. The structured mechanism-gap survey of Section 4.
5. The AV0-AV7 adversarial validation audit of Section 5, including the Julia bounded-equation-search engine through $C=20$.

8. Reproducibility

The public proof package bundles:

- `scan.py`, `canonical_bounded_prior.py`, `constants.json`, `benchmarks.json`, `declared_prior.csv`: the bounded 22,680-candidate scan, reproducible on a laptop in seconds.
- `top_candidates.csv`, `nist2026_top_candidates.csv`: the five nearest-neighbor tables of Section 2.
- `av6_engine/run_av6_exhaustive_c40_jl15.jl`, `av6_engine/launch_jl15.sh`, `av6_engine/spec.json`: the Julia bounded-equation-search engine source and launcher.
- `av6_outputs/c18/`, `av6_outputs/c19/`, `av6_outputs/c20/`: per-cost ledgers `bounded_shell_stats.csv`, `same_target_rank_tables.csv`, `sham_best_ledger.csv` directly auditable without re-running the engine. The $C = 18$ directory additionally contains the publication-grade KPI verdict file `diff_ledgers_publication_verdict.md`.
- `metrology_residuals.csv`, `mechanism_gap_routes.csv`: tabular sources of Sections 6 and 4.

- `manifest.json`: SHA-256 hashes for every package file plus the canonical candidate-list hash.
- `LICENSE`, `SCOPE.md`, `README.md`: reuse rights and explicit scope statement.

Three independent verification paths:

1. **Numerical.** Recompute G_{pred} from CODATA 2022 [1] values of α , m_e , $\hbar$, c , then verify the quoted residual and z -score.
2. **Algebraic.** Verify $\dim(\text{adj } SO(7)) = 21$, $h^\vee(B_3) = 5$, $\rho \cdot e_1 = 5/2$, and $\dim(SO(8))/\dim(SO(7)) = 28/21 = 4/3$ from B_3 root data [3].
3. **Bounded-statistical.** Run `python3 scan.py --benchmark codata2022 --limit 5;` expected output reproduces candidate-list SHA-256 `37ee945afb33e5c8f4a63b9874338fb38d8cab1e8b1d2c113873a2e632ff92f2` and the top-five table of Section 2. For the AV6 engine, `./av6_engine/launch_jl15.sh 18` reproduces the publication baseline in ~ 4.5 hours on 24 threads.

Statement of AI Authorship

This research and the resulting manuscript were carried out by an orchestration of AI agents under the named author’s direction. The author is an independent researcher without specialist training in theoretical physics. His role was vision, problem framing, target selection, and gate-keeping decisions: which leads to pursue, which to drop, when to escalate to multi-model adversarial review, and when a draft was acceptable. He did not personally derive the relation, identify the algebraic scaffold, design or operate the bounded-equation-search engine, or write the prose of this paper.

The technical work — empirical fit of G to the $\alpha^{21} \exp(-5\alpha/2)$ family; identification of the $B_3/SO(7)/SO(8)$ algebraic identities; design and execution of the bounded-prior reproducibility scan; implementation and runtime of the Julia bounded-equation-search engine and its sham-parity controls; structured survey of candidate mechanism routes; assembly of the public proof

package; and the writing of this manuscript — was performed by an orchestration of OpenAI GPT-5.4 / GPT-5.5, Google Gemini 3 Pro, Anthropic Claude Opus 4.7 and Sonnet 4.6, and Deepseek V4. Multi-model adversarial review served as the primary quality gate; council disagreements were surfaced to the named author for adjudication.

The named author chose the research target, accepted or rejected lines of work, approved every draft revision, and bears scientific responsibility for the content, claims, and citations herein.

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